



Efficient VLSI Timing Violation Screening: An Anomaly Detection Approach Using Lightweight 1D-CNNs

MTech Project Proposal Presentation



Project Summary

VLSI Physical Design is the stage in Integrated Chip Development in which logical design is converted into layout via Placement and Routing (PnR), keeping in mind the **Performance, Power and Area (PPA)** requirements. In this process **Static Timing Analysis** for timing verification has to be done repeatedly on each iteration of PnR optimization. Setup Timing Violation arises when a path fails timing verification, while Timing-Clean path is one that meets it

Repeated timing verifications required here becomes expensive. Graph Neural Networks (GNN) methods improve early timing estimation, but graph construction in them requires heavy training pipeline. This project proposes a lighter alternative using 1-dimensional Convolution Neural Networks (1-D CNN) for early risk screening rather than full slack regression. This is done by treating **Setup Timing Violations** as an anomaly detection problem, where violations are rare events among majority of the data being Timing-Clean paths.

Public OpenROAD datasets like EDA Schema V2 include essential information such as arrival time, slew, capacitance, etc. which we require for implementing this project. To address the heavy class imbalance (setup timing violations are rare), focal loss would be used and performance would be judged by metrics such as PR-AUC (area under the curve) and F1 score.



Objectives

1. To formulate pre-routing, post-placement setup timing violation screening as a path-level anomaly detection problem over structured sequential timing-path features derived from public OpenROAD datasets.
2. To design and implement a lightweight 1D-CNN model for early identification of likely timing-violating paths without performing full-graph slack regression.
3. To study focal-loss-based training strategies for severe class imbalance between timing-clean and timing-violating paths.



Literature Survey (Research papers refs in Proposal)

1. R. Jenila and K. Jagannadha Naidu, "A Survey on Learning-Based PnR Optimization for VLSI Physical Design Automation," *IEEE Access*, vol. 13, pp. 195953–195974, 2025. doi: 10.1109/access.2025.3631956.
2. T. Ajayi, V. Chhabria, M. Fogaça, S. Hashemi, A. Hosny, A. Kahng, M. Kim, J. Lee, U. Mallappa, M. Neseem, G. Pradipta, S. Reda, M. Saligane, S. Sapatnekar, C. Sechen, M. Shalan, W. Swartz, L. Wang, Z. Wang, M. Woo, and B. Xu, "INVITED: Toward an Open-Source Digital Flow: First Learnings from the OpenROAD Project," in *2019 56th ACM/IEEE Design Automation Conference (DAC)*, 2019, pp. 1–4. doi: 10.1145/3316781.3326334.
3. W. Ding, Z. Zhang, G. He, and P. Cao, "A Physical and Timing Aware Placement Optimization Framework Based on Graph Neural Network," in *2024 ACM/IEEE International Conference On Computer Aided Design (ICCAD)*, 2024, pp. 1–9. doi: 10.1145/3676536.3676772.
4. W. Gong, Y. Wang, M. Zhang, E. Mihankhah, H. Chen, and D. W. Wang, "A Fast Anomaly Diagnosis Approach Based on Modified CNN and Multisensor Data Fusion," *IEEE Transactions on Industrial Electronics*, vol. 69, pp. 13636–13646, 2022. doi: 10.1109/tie.2021.3135520.
5. Z. Guo, M. Liu, J. Gu, S. Zhang, D. Pan, and Y. Lin, "A Timing Engine Inspired Graph Neural Network Model for Pre-Routing Slack Prediction," in *2022 59th ACM/IEEE Design Automation Conference (DAC)*, 2022, pp. 1207–1212. doi: 10.1145/3489517.3530597.
6. A. Kahng and T. Spyrou, "The OpenROAD Project: Unleashing Hardware Innovation," 2021.
7. J. Li, J. Hu, Y. Wu, and X. Yang, "Pre-Routing Slack Prediction Based on Graph Attention Network," *Automation*, 2025. doi: 10.3390/automation6020020.
8. T.-Y. Lin, P. Goyal, R. B. Girshick, K. He, and P. Dollár, "Focal Loss for Dense Object Detection," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, pp. 318–327, 2017. doi: 10.1109/tpami.2018.2858826.
9. N. Mahmoodi, H. Shirazi, M. Fakhredanesh, and K. DadashtabarAhmadi, "Automatically weighted focal loss for imbalance learning," *Neural Computing and Applications*, vol. 37, pp. 4035–4052, 2024. doi: 10.1007/s00521-024-10323-x.
10. P. Shrestha, A. Aversa, and I. Savidis, "EDA-Schema-V2: A Multimodal Schema, Open Datasets, and Benchmarks for Machine Learning in Digital Physical Design," *arXiv preprint arXiv:2605.06952*, 2026. doi: 10.48550/arxiv.2605.06952.
11. P. Shrestha, A. Aversa, S. Phatharodom, and I. Savidis, *EDA-schema: A Graph Datamodel Schema and Open Dataset for Digital Design Automation*, Proceedings of the Great Lakes Symposium on VLSI 2024, 2024. doi: 10.1145/3649476.3658718.



Dataset

This project is planned around public-domain data from the OpenROAD ecosystem. The primary intended source is the EDA-Schema-V2 dataset. The dataset contains approximately 7,800 design instances across 18 benchmark circuits, stage-resolved representations from synthesis through detailed routing, and more than 36 million extracted timing paths. Importantly for this proposal, the dataset captures both timing-clean and timing-violating implementations generated through systematic sweeps of clock period, core utilization, and aspect ratio.

The public dataset link to EDA Schema V2 is <https://github.com/drexel-ice/EDA-schema>.



Workplan

1. **WP1: Problem formulation and data specification:** The first work package will define the anomaly-screening task at path level. The target will be a binary label indicating timing-clean versus timing-violating setup paths, consistent with the proposal scope. Sequential path records will be defined from timing-related attributes available in OpenROAD-derived data, focusing on features explicitly named in the project abstract such as arrival times, slew, and capacitance. The rationale is to replace full-netlist graph prediction with a lighter path-sequence representation.
2. **WP2: Dataset extraction and preprocessing.** Public OpenROAD-derived datasets described in EDA-Schema-V2 will be inspected to identify the exact stage-resolved data suitable for pre-routing, post-placement timing-risk screening. Candidate designs will be selected from the publicly released benchmark suite, and path samples will be extracted, normalized, and padded or truncated to a fixed sequence format for 1D-CNN input. Because the dataset includes both timing-clean and timing-violating implementations generated by systematic parameter sweeps, it supports controlled sampling for anomaly detection experiments \cite{Shrestha2026EDA-Schema-V2:}.
3. **WP3: Lightweight 1D-CNN design.** A compact 1D-CNN classifier will be implemented for binary anomaly screening. The design rationale is that 1D-CNNs are well suited to raw or structured 1-D sequential inputs and can reduce parameter count relative to heavier alternatives \cite{Gong2022A}. The model will remain intentionally lightweight so that training and inference are feasible on consumer hardware.
4. **WP4: Imbalance-aware training and evaluation.** Training will compare standard binary cross-entropy against focal-loss-based objectives. The rationale is that focal loss reduces domination by easy majority examples and emphasizes hard minority cases \cite{Lin2017Focal}. Experiments will evaluate PR-AUC and F1-score as primary metrics, since the target class is rare. Precision, recall, confusion matrix, and training/inference time will be collected as supporting metrics.
5. **WP5: Baseline comparison and resource study.** The final work package will compare the proposed method against literature-reported graph-based timing predictors qualitatively on task formulation and quantitatively on screening performance and computational footprint where feasible. Because recent GNN timing systems use deeper graph models and stronger hardware \cite{Ding2024A}, this package will emphasize efficiency-oriented comparison: model size, memory demand, CPU feasibility, and optional low-cost GPU usage.



Expected Deliverables and Implementation Arrangements

The primary outcome of this project is the development and validation of a lightweight 1D-CNN framework designed for rapid, pre-routing setup timing violation screening in VLSI physical design.

Planned Implementation Details are:

- **Lightweight Screening Pipeline:** PyTorch-based 1D-CNN framework and data preprocessing pipeline for fast, path-level setup timing violation screening using public OpenROAD datasets.
- **Imbalance-Aware Model & Benchmarks:** Focal-loss training strategy tailored for severe class imbalance, evaluated via PR-AUC, F1-score, precision, and recall.
- **Efficiency & Resource Report:** Quantitative benchmark covering model size, inference speed, and compute feasibility on personal CPU/GPU hardware.



Resource Requirements

Software: Python, PyTorch, common scientific-computing libraries, and data-processing tools are required. These software components are open source and publicly accessible.

Hardware: The project is intended to be feasible on a laptop and cloud notebook GPUs for data preparation, preprocessing, experiments, and baseline model development.